

Gamma funⁿ.

The funⁿ of n . defined as.

$$\int_0^{\infty} e^{-x} x^{n+1} dx = \Gamma n \quad (\text{when } n > 0)$$

is called Gamma funⁿ.

$$\boxed{\int_0^{\infty} e^{-x} x^n dx = \Gamma(n+1)}$$

* Properties of gamma funⁿ

$$\textcircled{1} \Gamma 1 = 1$$

$$\Gamma 0, \Gamma n = \infty$$

$\hookrightarrow n$ is -ve

$$\textcircled{2} \Gamma \frac{1}{2} = \sqrt{\pi}$$

$$\textcircled{3} \Gamma p \Gamma(1-p) = \frac{\pi}{\sin p\pi} \quad 0 < p < 1$$

$$\text{eg } \Gamma \frac{1}{4} \Gamma \frac{3}{4} = \Gamma \frac{1}{4} \Gamma 1 - \frac{1}{4} = \frac{\pi}{\sin \pi/4} = \sqrt{2}\pi$$

$$\textcircled{4} \Gamma n = (n-1) \Gamma(n-1)$$

$$\text{eg } \Gamma \frac{5}{2} = \frac{3}{2} \Gamma \frac{3}{2} = \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2} = \frac{3\sqrt{\pi}}{4}$$

$$\textcircled{5} \Gamma n = (n-1)! \quad \text{if } n \text{ is +ve int.}$$

$$\begin{aligned} \Gamma 4 &= 3 \Gamma 3 \\ &= 3 \cdot 2 \Gamma 2 = 3 \cdot 2 \cdot 1 \Gamma 1 \\ &= 3! \end{aligned}$$

$$I = \int_0^{\infty} e^{-a^2 x^2} dx$$

$$\int_0^{\infty} e^{-x} x^n dx = \sqrt{n+1}$$

$$a^2 x^2 = t \quad x^2 = \frac{t}{a^2}$$

$$x = \frac{t^{1/2}}{a} \Rightarrow dx = \frac{1}{2a} t^{-1/2} dt$$

$$x: 0 \rightarrow \infty$$

$$t: 0 \rightarrow \infty$$

$$= \int_0^{\infty} e^{-t} \frac{1}{2a} t^{-1/2} dt = \frac{1}{2a} \int_0^{\infty} e^{-t} t^{-1/2} dt$$

$$= \frac{1}{2a} \sqrt{-\frac{1}{2} + 1} = \frac{1}{2a} \sqrt{\frac{1}{2}} = \frac{\sqrt{\pi}}{2a}$$

$$\int_0^{\infty} x^{1/4} e^{-\sqrt{x}} dx$$

$$t^{1/2} +$$



$$\sqrt{x} = t \Rightarrow x = t^2$$

$$dx = 2t dt$$

$$x: 0 \rightarrow \infty \quad t: 0 \rightarrow \infty$$

$$= \int_0^{\infty} (t^2)^{1/4} e^{-t} 2t dt$$

$$= 2 \int_0^{\infty} e^{-t} t^{3/2} dt$$

$$= 2 \sqrt{\frac{3}{2} + 1} = 2 \sqrt{\frac{5}{2}}$$

$$= \sqrt{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}}$$

$$= \frac{3\sqrt{\pi}}{2} //$$

$$\Gamma n = (n-1) \Gamma (n-1)$$

$$p.p. - \int_0^\infty x e^{-x^8} dx \cdot \int_0^\infty x^2 e^{-x^4} dx = \frac{\pi}{16\sqrt{2}}$$

$$I_1 = \int_0^\infty x e^{-x^8} dx$$

$$x^8 = t$$

$$x = t^{1/8}$$

$$dx = \frac{1}{8} t^{-7/8} dt$$

$$x: 0 \rightarrow \infty \quad t: 0 \rightarrow \infty$$

$$= \int_0^\infty (t)^{1/8} e^{-t} \frac{1}{8} t^{-7/8} dt$$

$$= \frac{1}{8} \int_0^\infty e^{-t} t^{-6/8} dt$$

$$= \frac{1}{8} \Gamma\left(\frac{1}{4}\right)$$

$$\int_0^\infty x e^{-x^8} dx \cdot \int_0^\infty x e^{-x^4} dx = \frac{1}{8} \Gamma\left(\frac{1}{4}\right) \frac{1}{4} \Gamma\left(\frac{3}{4}\right) =$$

$$= \frac{1}{32} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)$$

$$= \frac{1}{32} \Gamma\left(\frac{1}{4}\right) \Gamma\left(1 - \frac{1}{4}\right)$$

$$= \frac{1}{32} \frac{\pi}{\sin \pi/4} = \frac{1}{32} \sqrt{2} \pi \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\pi}{16\sqrt{2}}$$

$$I_2 = \int_0^\infty x^2 e^{-x^4} dx$$

$$x^4 = t$$

$$x = t^{1/4}$$

$$dx = \frac{1}{4} t^{-3/4} dt$$

$$= \int_0^\infty (t^{1/4})^2 e^{-t} \frac{1}{4} t^{-3/4} dt$$

$$= \frac{1}{4} \int_0^\infty e^{-t} t^{1/4} dt$$

$$= \frac{1}{4} \Gamma\left(\frac{5}{4}\right)$$

$$\int x^m (\log \frac{1}{x})^n dx$$



$$\log \frac{1}{x} = t$$

$$\Rightarrow \frac{1}{x} = e^t$$

$$\Rightarrow x = e^{-t}$$

$$dx = -e^{-t} dt$$

$$x: 0 \rightarrow 1 \quad t: \infty \rightarrow 0$$

$$I = \int_0^1 (e^{-t})^m (t)^n (-e^{-t}) dt$$

$$= \int_0^{\infty} e^{-(m+1)t} t^n dt$$

$$(m+1)t = y \Rightarrow t = \frac{y}{m+1}$$

$$dt = \frac{dy}{m+1}$$

$$t: 0 \rightarrow \infty$$

$$y: 0 \rightarrow \infty$$

$$= \int_0^{\infty} e^{-y} \left(\frac{y}{m+1}\right)^n \frac{dy}{m+1}$$

$$= \frac{1}{(m+1)^{n+1}} \int_0^{\infty} e^{-y} y^n dy$$

$$= \frac{n!}{(m+1)^{n+1}}$$

(Hw)

$$\int (x \log x)^3 dx$$

$$\log x = -t$$

$$\Rightarrow x = e^{-t}$$

(36)

$$\int_0^{\infty} \frac{x^7}{7^x} dx = \int_0^{\infty} x^7 \left(7^{-x} \right) dx$$

$$\rightarrow 7^{-x} = e^{-t} \Rightarrow \log 7^{-x} = \log e^{-t}$$

$$+ x \log 7 = -t$$

$$x = \frac{-t}{\log 7} \Rightarrow dx = \frac{-dt}{\log 7}$$

$$x: 0 \rightarrow \infty \quad -t: 0 \rightarrow \infty$$

$$I = \int_0^{\infty} \left(\frac{-t}{\log 7} \right)^7 e^{-t} \frac{-dt}{\log 7}$$

$$= \frac{1}{(\log 7)^8} \int_0^{\infty} e^{-t} t^7 dt = \frac{1}{(\log 7)^8} \Gamma(8) = \frac{7!}{(\log 7)^8}$$

$$\int_0^{\infty} 7^{-4x^2} dx$$

$$\rightarrow 7^{-4x^2} = e^{-t}$$

$$+ 4x^2 \log 7 = -t$$

$$x = \frac{-t^{1/2}}{2(\log 7)^{1/2}}$$

$$dx = \frac{1}{4} \frac{-t^{-1/2}}{(\log 7)^{1/2}} dt$$

$$x: 0 \rightarrow \infty \quad t: 0 \rightarrow \infty$$

$$I = \int_0^{\infty} e^{-t} \frac{1}{4} \frac{t^{-1/2}}{(\log 7)^{1/2}} dt = \frac{1}{4\sqrt{\log 7}} \int_0^{\infty} e^{-t} t^{-1/2} dt$$

$$= \frac{1}{4\sqrt{\log 7}} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{4\sqrt{\log 7}}$$

$$\int_0^{\infty} e^{-x} x^n dx = \Gamma(n+1)$$

$$\Gamma n = (n-1) \Gamma(n-1)$$

$$\Gamma n = (n-1)!$$

$$\Gamma \frac{1}{2} = \sqrt{\pi}$$

$$\Gamma p \Gamma(1-p) = \frac{\pi}{\sin p\pi}$$

$$0 < p < 1$$

give $\sqrt{1.8} = 0.9314$ find $\sqrt{-2.2}$

$$\sqrt{n} = (n-1) \sqrt{n-1}$$

$$\left. \begin{aligned} \sqrt{n+1} &= n \sqrt{n} \\ \sqrt{n} &= \frac{\sqrt{n+1}}{n} \end{aligned} \right\}$$

$$\sqrt{-2.2} = \frac{\sqrt{-1.2}}{(-2.2)} = \frac{\sqrt{-0.2}}{(-2.2)(-1.2)} = \frac{\sqrt{0.8}}{(2.2)(1.2)(-0.2)}$$

$$= - \frac{\sqrt{1.8}}{(2.2)(1.2)(0.2)(0.8)}$$

$$= - \frac{0.9314}{(2.2)(1.2)(0.2)(0.8)}$$

$$= -2.205 \approx -2.21$$

P.T

$$\sqrt{n} = (n-1) \sqrt{n-1} \quad \underline{\text{Inductive}}$$

$$\sqrt{n+1/2} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n} \sqrt{\pi}$$

$$\sqrt{n+1/2} = (n-1/2) \sqrt{n-1/2}$$

$$= (n-1/2) (n-3/2) \sqrt{n-3/2}$$

$$= (n-1/2) (n-3/2) \cdots \frac{5}{2} \frac{3}{2} \frac{1}{2} \sqrt{1/2}$$

$$= \frac{(2n-1)}{2} \frac{(2n-3)}{2} \cdots \frac{5}{2} \frac{3}{2} \frac{1}{2} \sqrt{\pi}$$

$$= \frac{1 \cdot 3 \cdot 5 \cdots (2n-3)(2n-1)}{2^n} \sqrt{\pi}$$

$$I_n = \frac{\frac{\sqrt{\pi}}{2} \sqrt{\frac{n+1}{2}}}{\sqrt{\frac{n}{2} + 1}}$$

Show that $I_{n+2} = \frac{n+1}{n+2} I_n$

$$I_{n+2} = \frac{\frac{\sqrt{\pi}}{2} \sqrt{\frac{n+2+1}{2}}}{\sqrt{\frac{n+2}{2} + 1}} = \frac{\frac{\sqrt{\pi}}{2} \sqrt{\frac{n+3}{2}}}{\sqrt{\frac{n+2}{2} + 1}} = \frac{\frac{\sqrt{\pi}}{2} \frac{n+1}{2} \sqrt{\frac{n+1}{2}}}{\frac{n+2}{2} \sqrt{\frac{n+2}{2}}}$$

$$\sqrt{n} = (n-1) \sqrt{n-1}$$

$$= \frac{n+1}{n+2} \frac{\frac{\sqrt{\pi}}{2} \sqrt{\frac{n+1}{2}}}{\sqrt{\frac{n}{2} + 1}}$$

$$= \frac{n+1}{n+2} I_n$$

Beta function

for $m, n > 0$ beta funⁿ is defined.

$$\int_0^1 x^{m-1} (1-x)^{n-1} dx = \beta(m, n)$$

$$\int_0^1 x^m (1-x)^n dx = \beta(m+1, n+1)$$

Types

$$\begin{aligned} \textcircled{1} \quad & \int_0^a x^m (a-x)^n dx \\ &= a^n \int_0^a x^m \left(1 - \frac{x}{a}\right)^n dx \end{aligned}$$

$$\begin{aligned} \text{put } \frac{x}{a} &= t \\ x &= at \\ dx &= a dt \end{aligned}$$

$$x: 0 \rightarrow a$$

$$t: 0 \rightarrow 1$$

$$\textcircled{2} \quad \int_0^1 x^m (1-x^n)^p dx$$

$$\begin{aligned} \text{put } x^n &= t \\ x &= t^{1/n} \\ dx &= \frac{1}{n} t^{1/n-1} dt \end{aligned} \quad \begin{aligned} x: 0 &\rightarrow 1 \\ t: 0 &\rightarrow 1 \end{aligned}$$

$$\textcircled{3} \quad \int_a^b (x-a)^m (b-x)^n dx$$

$$x = a + (b-a)t$$

$$x-a = (b-a)t$$

$$x: a \rightarrow b$$

$$t: 0 \rightarrow 1$$

Properties . .

$$(1) \quad \beta(m, n) = \beta(n, m)$$

$$(2) \quad \beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$$

$$(3) \quad \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \beta(m, n)$$

$$(4) \quad \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$p = q = 0$$

$$\int_0^{\pi/2} d\theta = \frac{1}{2} \beta\left(\frac{1}{2}, \frac{1}{2}\right)$$

$$\frac{\pi}{2} = \frac{1}{2} \frac{\Gamma_{1/2} \Gamma_{1/2}}{\Gamma_1} \Rightarrow \pi = (\Gamma_{1/2})^2$$
$$\Gamma_{1/2} = \sqrt{\pi}$$

(5) Duplication formula

$$2^{2m-1} \Gamma m \Gamma m + \frac{1}{2} = \sqrt{\pi} \Gamma 2m$$

$$I = \int_0^1 \sqrt{1-\sqrt{x}} \, dx \quad \int_0^{1/2} \sqrt{2y-4y^2} \, dy$$

$$I_1 = \int_0^1 \sqrt{1-\sqrt{x}} \, dx$$

$$= \int_0^1 (1-\sqrt{x})^{1/2} \, dx$$

$$\sqrt{x} = t$$

$$\Rightarrow x = t^2$$

$$dx = 2t \, dt$$

$$x: 0 \rightarrow 1 \quad t: 0 \rightarrow 1$$

$$= \int_0^1 (1-t)^{1/2} 2t \, dt$$

$$= 2 \int_0^1 t(1-t)^{1/2} \, dt$$

$$= 2 \, \beta(2, 3/2)$$

$$I_2 = \int_0^{1/2} \sqrt{2y-4y^2} \, dy$$

$$= \int_0^{1/2} (2y)^{1/2} (1-2y)^{1/2} \, dy$$

$$2y = t$$

$$y = t/2$$

$$dy = \frac{dt}{2}$$

$$y: 0 \rightarrow 1/2 \quad t: 0 \rightarrow 1$$

$$= \sqrt{2} \int_0^1 \left(\frac{t}{2}\right)^{1/2} (1-t)^{1/2} \frac{dt}{2}$$

$$= \frac{\sqrt{2}}{\sqrt{2} \cdot 2} \int_0^1 t^{1/2} (1-t)^{1/2} \, dt$$

$$= \frac{1}{2} \, \beta(3/2, 3/2)$$

$$I = I_1 - I_2 = 2 \, \beta(2, 3/2) - \frac{1}{2} \, \beta(3/2, 3/2)$$

$$= \frac{\sqrt{2} \, \Gamma_{3/2}}{\Gamma_{7/2}} \frac{\Gamma_{3/2} \Gamma_{3/2}}{\Gamma_3}$$

$$= \frac{1! \sqrt{3/2}}{\frac{5}{2} \sqrt{3/2} \sqrt{3/2}} \frac{(\sqrt{3/2})^2}{2!}$$

$$= \frac{4}{15} \frac{(\frac{1}{2} \Gamma_{1/2})^2}{2} = \frac{4}{15} \times \frac{1}{4} \frac{(\sqrt{\pi})^2}{2}$$

$$= \frac{\pi}{30} //$$

$$\int_0^1 x^m (1-x)^n dx = \beta(m+1, n+1)$$

$$\int_0^{2a} x^2 \sqrt{2ax - x^2} dx$$

$$= \int_0^{2a} x^2 \sqrt{2ax} \left(1 - \frac{x}{2a}\right)^{1/2} dx$$

$$= \sqrt{2a} \int_0^{2a} x^{5/2} \left(1 - \frac{x}{2a}\right)^{1/2} dx \quad \begin{matrix} \frac{x}{2a} = t & x: 0 \rightarrow 2a \\ t: 0 \rightarrow 1 \end{matrix}$$

$$= \sqrt{2a} \int_0^1 (2at)^{5/2} (1-t)^{1/2} 2a dt \quad \begin{matrix} x = 2at \\ dx = 2a dt \end{matrix}$$

$$= (2a)^{7/2} 2a \int_0^1 t^{5/2} (1-t)^{1/2} dt$$

$$= (2a)^4 \frac{\beta\left(\frac{7}{2}, \frac{3}{2}\right)}{\frac{\Gamma(7/2) \Gamma(3/2)}{\Gamma(5)}} = \frac{16a^4 \frac{5}{2} \frac{3}{2} \frac{1}{2} \frac{\Gamma(1/2)}{2} \frac{\Gamma(1/2)}{2}}{\frac{4!}{(n-(n-1)) \Gamma(n-1)}}$$

$$= \frac{16a^4 \frac{15}{8} \pi}{15 \cdot 24} = \frac{a^4 5\pi}{8}$$

p. 7.

$$\int \frac{dx}{\sqrt{1-x^n}} = \frac{\int \frac{dx}{\sqrt{1-x^n}}}{\frac{1}{\sqrt{\frac{1}{n}}}}$$

$$I = \int_0^1 (1-x^n)^{-1/2} dx \quad \begin{matrix} x^n = t \\ x = t^{1/n} \\ dx = \frac{1}{n} t^{1/n-1} dt \end{matrix} \quad \begin{matrix} x: 0 \rightarrow 1 \\ t: 0 \rightarrow 1 \end{matrix}$$

$$= \int_0^1 (1-t)^{-1/2} \frac{1}{n} t^{1/n-1} dt$$

$$= \frac{1}{n} \int_0^1 t^{1/n-1} (1-t)^{-1/2} dt = \frac{1}{n} \beta\left(\frac{1}{n}, \frac{1}{2}\right)$$

$$\text{By Duplication formula} \quad \frac{1}{n} \frac{\Gamma(1/n) \Gamma(1/2)}{\Gamma(1/n + 1/2)} = \frac{1}{n} \frac{\Gamma(1/n) \sqrt{\pi}}{\Gamma(1/n + 1/2)}$$

$$2^{2m-1} \Gamma(m) \Gamma(m+1/2) = \sqrt{\pi} \Gamma(2m)$$

$$\text{put } m = \frac{1}{n} \quad 2^{\frac{2}{n}-1} \Gamma\left(\frac{1}{n}\right) \Gamma\left(\frac{1}{n} + \frac{1}{2}\right) = \sqrt{\pi} \Gamma\left(\frac{2}{n}\right) \Rightarrow \Gamma\left(\frac{1}{n} + \frac{1}{2}\right) = \frac{\sqrt{\pi} \Gamma\left(\frac{2}{n}\right)}{2^{\frac{2}{n}-1} \Gamma\left(\frac{1}{n}\right)}$$

$$\int_0^1 \frac{dx}{\sqrt{1-x^n}} = \frac{\frac{1}{n} \Gamma(1/n) \sqrt{\pi}}{\sqrt{\pi} \Gamma(2/n)} 2^{\frac{2}{n}-1} \Gamma(2/n) = \frac{(\Gamma(1/n))^2 2^{\frac{2}{n}-1}}{n \Gamma(2/n)}$$

$$\int_0^a x^4 (a^2 - x^2)^{1/2} dx = \frac{a^6 \pi}{32}$$

$$= a \int_0^a x^4 \left(1 - \frac{x^2}{a^2}\right)^{1/2} dx \quad \begin{matrix} \frac{x^2}{a^2} = t \Rightarrow x = at \\ dx = \frac{a}{2} t^{-1/2} dt \end{matrix}$$

$$= a \int_0^1 (at)^4 (1-t)^{1/2} \frac{a}{2} t^{-1/2} dt \quad \begin{matrix} x: 0 \rightarrow a \\ t: 0 \rightarrow 1 \end{matrix}$$

$$= \frac{a^6}{2} \int_0^1 t^{3/2} (1-t)^{1/2} dt = \frac{a^6}{2} \beta\left(\frac{5}{2}, \frac{3}{2}\right)$$

$$= \frac{a^6}{2} \frac{\Gamma(5/2) \Gamma(3/2)}{\Gamma(4)} = \frac{a^6 \pi}{32}$$

(I)

$$\int \frac{x^7}{\sqrt{1-x^2}} dx.$$

put $x^2 = t$

(II)

put $x = \sin \theta$

$dx = \cos \theta d\theta$

$x: 0 \rightarrow 1$

$\theta: 0 \rightarrow \pi/2$

$$I = \int_0^{\pi/2} \frac{\sin^7 \theta \cos \theta d\theta}{\sqrt{1-\sin^2 \theta}}$$

$$= \int_0^{\pi/2} \frac{\sin^7 \theta \cos \theta d\theta}{\cancel{\cos \theta}}$$

$$\Gamma n = (n-1) \Gamma n-1$$

$$\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta$$

$$= \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$p=7, q=0$

$$= \frac{1}{2} B\left(\frac{8}{2}, \frac{1}{2}\right)$$

$$= \frac{1}{2} \frac{\Gamma 4 \Gamma \frac{1}{2}}{\Gamma \frac{9}{2}}$$

$$= \frac{1}{2} \frac{3! \sqrt{\pi}}{\frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}}$$

$$= \frac{16 \times 6^2}{2 \times 7 \times 5 \times 3} = \frac{16}{35}$$

$$= \int_0^1 \frac{x^2(4-x^4)}{\sqrt{1-x^2}} dx$$

$x = \sin \theta$
 $dx = \cos \theta d\theta$
 $x: 0 \rightarrow 1$
 $\theta: 0 \rightarrow \pi/2$

$$= \int_0^{\pi/2} \frac{\sin^2 \theta (4 - \sin^4 \theta) \cancel{\cos \theta} d\theta}{\cancel{\cos \theta}}$$

$$= \int_0^{\pi/2} 4 \sin^2 \theta d\theta - \int_0^{\pi/2} \sin^6 \theta d\theta$$

$$= \frac{4}{2} \beta\left(\frac{3}{2}, \frac{1}{2}\right) - \frac{1}{2} \beta\left(\frac{7}{2}, \frac{1}{2}\right)$$

rw

$$= \frac{2\pi}{32}$$

$$\int_0^{\pi} \frac{\sin^4 \theta}{(1 + \cos \theta)^2} d\theta \quad 1 + \cos 2\theta = 2 \cos^2 \theta$$

$$= \int_0^{\pi} \frac{\sin^4 \theta}{(2 \cos^2 \theta/2)^2} d\theta$$

$$= \int_0^{\pi} \frac{(2 \sin \theta/2 \cos \theta/2)^4}{4 \cos^4 \theta/2} d\theta$$

$$= \frac{16}{4} \int_0^{\pi} \sin^4 \theta/2 d\theta$$

$$= 4 \int_0^{\pi/2} \sin^4 t \cdot 2 dt$$

$$= \frac{8}{2} \beta\left(\frac{5}{2}, \frac{1}{2}\right)$$

$$= 4 \frac{\Gamma_{\frac{5}{2}} \Gamma_{\frac{1}{2}}}{\Gamma_3} = \frac{4 \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma_{\frac{1}{2}} \Gamma_{\frac{1}{2}}}{2} = \frac{3\pi}{2}$$

$$\begin{aligned} \theta/2 &= t & \theta: 0 \rightarrow \pi \\ t: 0 &\rightarrow \pi/2 \\ \theta &= 2t \\ d\theta &= 2dt \end{aligned}$$

$$\int_0^{\pi/6} \cos^3 3\theta \sin^2 6\theta d\theta.$$

$$\begin{aligned} 3\theta &= t \Rightarrow \theta = t/3 & d\theta &= \frac{dt}{3} \\ \theta: 0 &\rightarrow \pi/6 & t: 0 &\rightarrow \pi/2 \end{aligned}$$

$$= \int_0^{\pi/2} \cos^3 t \sin^2 2t \frac{dt}{3}$$

$$= \frac{1}{3} \int_0^{\pi/2} \cos^3 t (2 \sin t \cos t)^2 dt$$

$$= \frac{4}{3} \int_0^{\pi/2} \sin^2 t \cos^5 t dt = \frac{4}{3} \cdot \frac{1}{2} \beta\left(\frac{3}{2}, \frac{6}{2}\right)$$

$$\left(\frac{32}{315} \right)$$

Ans

$$\int_0^1 x^4 \cos^{-1} x \, dx$$

U(1)ATE

$$= \int_0^1 \underbrace{\cos^{-1} x}_u \underbrace{x^4}_v \, dx$$

$$= \left[\cos^{-1} x \cdot \frac{x^5}{5} - \int \frac{x^5}{5} \left(\frac{-1}{\sqrt{1-x^2}} \right) dx \right]_0^1$$

$$= \left[\cos^{-1} x \cdot \frac{x^5}{5} \right]_0^1 + \frac{1}{5} \int_0^1 \frac{x^5}{\sqrt{1-x^2}} \, dx$$

$$= [0 - 0] + \frac{1}{5} \int_0^1 \frac{x^5}{\sqrt{1-x^2}} \, dx$$

put $x = \sin \theta$
 $dx = \cos \theta \, d\theta$ $\theta: 0 \rightarrow \frac{\pi}{2}$

$$= \frac{1}{5} \int_0^{\pi/2} \frac{\sin^5 \theta \cos \theta}{\cos \theta} \, d\theta$$

$$= \frac{1}{5} \times \frac{1}{2} B\left(\frac{6}{2}, \frac{1}{2}\right)$$

$$= \frac{8}{75}$$

$$\begin{aligned}
& \int_0^{2\pi} \sin^2 \theta (1 + \cos \theta)^2 d\theta \\
&= \int_0^{2\pi} \sin^2 \theta (2 \cos^2 \theta/2)^2 d\theta \\
&= \int_0^{2\pi} (2 \sin \theta/2 \cos \theta/2)^2 + \cos^4 \theta/2 d\theta \\
&= 16 \int_0^{2\pi} \sin^2 \theta/2 \cos^6 \theta/2 d\theta
\end{aligned}$$

$$\frac{\theta}{2} = t \Rightarrow \theta = 2t \\
d\theta = 2 dt$$

$$\begin{aligned}
\theta: 0 &\rightarrow 2\pi \\
t: 0 &\rightarrow \pi
\end{aligned}$$

$$= 16 \int_0^{\pi} \sin^2 t \cos^6 t \cdot 2 dt$$

$$= 32 \int_0^{\pi} \sin^2 t \cos^6 t dt$$

$$\left. \begin{aligned} \int_0^{2a} f(t) dt &= 2 \int_0^a f(t) dt \\ \text{if } f(\pi-t) &= f(t) \end{aligned} \right\}$$

$$\left. \begin{aligned} \sin(\pi-t) &= -\cos \pi \sin t = \sin t \\ \cos(\pi-t) &= -\cos t \end{aligned} \right\}$$

$$f(t) = \sin^2 t \cos^6 t$$

$$f(\pi-t) = \sin^2(\pi-t) \cos^6(\pi-t) = \sin^2 t \cos^6 t = f(t)$$

$$I = 32 \times 2 \int_0^{\pi/2} \sin^2 t \cos^6 t dt = \frac{32 \times 2}{2} B\left(\frac{3}{2}, \frac{7}{2}\right)$$

$$= \left(\right)$$

5TH 421.

Evaluate $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta$ & p-T.

$$\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \frac{\pi}{2}$$

$$\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \int_0^{\pi/2} \sin^{1/2} \theta \cos^{-1/2} \theta d\theta$$

$$= \frac{1}{2} B\left(\frac{3}{4}, \frac{1}{4}\right)$$

$$= \frac{1}{2} \frac{\sqrt{\frac{3}{4}} \sqrt{\frac{1}{4}}}{\sqrt{\pi}} = \frac{1}{2} \frac{\sqrt{2}\pi}{\sqrt{2}} = \frac{\pi}{2}$$

$$\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad 0 < p < 1$$

$$\sqrt{p} \sqrt{1-p} = \frac{\pi}{8 \sin \pi}$$

$$\sqrt{\frac{3}{4}} \sqrt{\frac{1}{4}} = \sqrt{2}\pi$$

$$\int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \int_0^{\pi/2} \sin^{-1/2} \theta \cos^{1/2} \theta d\theta = \frac{1}{2} B\left(\frac{1}{4}, \frac{3}{4}\right)$$

$$= \frac{\pi}{2}$$

$$\int_{-\pi/4}^{\pi/4} (\sin \theta + \cos \theta)^{1/3} d\theta.$$

$$= \int_{-\pi/4}^{\pi/4} \left[\sqrt{2} \left(\sin \theta \frac{1}{\sqrt{2}} + \cos \theta \frac{1}{\sqrt{2}} \right) \right]^{1/3} d\theta$$

$$= \int_{-\pi/4}^{\pi/4} 2^{1/6} \left(\sin \theta \cos \pi/4 + \cos \theta \sin \pi/4 \right)^{1/3} d\theta$$

$$= 2^{1/6} \int_{-\pi/4}^{\pi/4} \left[\sin \left(\theta + \pi/4 \right) \right]^{1/3} d\theta.$$

$$= 2^{1/6} \int_0^{\pi/2} \sin^{1/3} t dt$$

$$= 2^{1/6} \frac{1}{2} B \left(\frac{4}{3}, \frac{1}{2} \right)$$

$$= \frac{2^{-5/6} \sqrt{2/3} \Gamma(1/2)}{\Gamma(7/6)} = \frac{2^{-5/6} \sqrt{2/3} \sqrt{\pi}}{\Gamma(7/6)}$$

$$\theta + \pi/4 = t$$

$$d\theta = dt$$

$$\theta: -\pi/4 \rightarrow \pi/4$$

$$t: 0 \rightarrow \pi/2$$

$$\int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \beta(m, n)$$

$$\int_0^{\infty} \frac{x^2}{(1+x)^3} dx$$

$$= \int_0^{\infty} \frac{(t^{1/6})^2}{(1+t)^3} \cdot \frac{1}{6} t^{-5/6} dt$$

$$= \frac{1}{6} \int_0^{\infty} \frac{t^{-3/6}}{(1+t)^3} dt$$

$$= \frac{1}{6} \beta\left(\frac{1}{2}, \frac{5}{2}\right)$$

$$= \frac{1}{6} \frac{\Gamma_{1/2} \Gamma_{5/2}}{\Gamma_3} = \frac{1}{6} \frac{\sqrt{\pi}^{3/2} \frac{1}{2} \Gamma_{1/2}}{2!} = \frac{1 \times 3! \cdot \pi^{5/2}}{2 \times 8} = \frac{\pi}{16}$$

$$\int_0^{\infty} \frac{\sqrt{x}}{(1+2x+x^2)} dx =$$

$$\int_0^{\infty} \frac{x^{1/2}}{(1+x)^2} dx$$

$$m-1 = \frac{1}{2}$$

$$m+n = 2$$

$$x^6 = t$$

$$x = t^{1/6}$$

$$dx = \frac{1}{6} t^{-5/6} dt$$

$$x: 0 \rightarrow \infty$$

$$t: 0 \rightarrow \infty$$

$$m-1 = -\frac{1}{2} \Rightarrow m = \frac{1}{2}$$

$$m+n = 3 \Rightarrow n = 3 - \frac{1}{2}$$

P.T $\int_0^\infty \frac{x^{m-1}}{(a+bx)^{m+n}} dx = \frac{1}{a^n b^m} \beta(m, n)$

& Hence And $\int_0^\infty \frac{\sqrt{x}}{(4+4x+x^2)} dx$

$I = \frac{1}{a^{m+n}} \int_0^\infty \frac{x^{m-1}}{(1+\frac{b}{a}x)^{m+n}} dx$

$\frac{b}{a}x = \tan^2 \theta$

$x = \frac{a}{b} \tan^2 \theta \Rightarrow dx = \frac{a}{b} 2 \tan \theta \sec^2 \theta d\theta$

$x: 0 \rightarrow \infty \quad \theta: 0 \rightarrow \pi/2$

$= \frac{1}{a^{m+n}} \int_0^{\pi/2} \frac{\left(\frac{a}{b} \tan^2 \theta\right)^{m-1}}{(1+\tan^2 \theta)^{m+n}} \frac{a}{b} 2 \tan \theta \sec^2 \theta d\theta$

$= \frac{2}{a^{m+n}} \frac{a^{m-1}}{b^{m-1}} \frac{a}{b} \int_0^{\pi/2} \frac{\tan^{2m-2} \theta}{(\sec^2 \theta)^{m+n}} \tan \theta \sec^2 \theta d\theta$

$= \frac{2}{a^n b^m} \int_0^{\pi/2} \frac{\tan^{2m-1} \theta}{\sec^{2m+2n-2} \theta} d\theta$

$= \frac{2}{a^n b^m} \int_0^{\pi/2} \frac{\sin^{2m-1} \theta}{\cos^{2m+1} \theta} (\cos \theta)^{2m+2n-2} d\theta$

$= \frac{2}{a^n b^m} \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2m+2n-2-2m+1} \theta d\theta$

$= \frac{2}{a^n b^m} \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

$\int_0^\infty \frac{x^{m-1}}{(a+bx)^{m+n}} dx = \frac{2}{a^n b^m} \frac{1}{2} \beta(m, n)$

$\int_0^\infty \frac{\sqrt{x}}{(4+4x+x^2)} dx$
 $= \int_0^\infty \frac{x^{1/2}}{(2+x)^2} dx$

$a=2, b=1$

$m-1 = \frac{1}{2} \quad m+n=2$

$$= \int_3^7 \sqrt[4]{(x-3)(7-x)} \, dx$$

$$= \int_3^7 (x-3)^{1/4} (7-x)^{1/4} \, dx$$

$$x-3 = \underset{\substack{\uparrow \\ (7-3)}}{4} t$$

$$dx = 4 \, dt$$

$$x: 3 \rightarrow 7$$

$$t: 0 \rightarrow 1$$

$$= \int_0^1 (4t)^{1/4} (7-3-4t)^{1/4} \, dt$$

$$= 4^{1/4} \cdot 4 \int_0^1 t^{1/4} (4-4t)^{1/4} \, dt$$

$$= 4^{1/4} \cdot 4^{1/4} \cdot 4 \int_0^1 t^{1/4} (1-t)^{1/4} \, dt = 4^{1/2} \cdot 4 \cdot \beta\left(\frac{5}{4}, \frac{5}{4}\right)$$

$$= 2 \times 4 \cdot \frac{(\Gamma(5/4))^2}{\Gamma(5/2)} = \frac{8 \left(\frac{1}{4} \Gamma(1/4)\right)^2}{\frac{3}{2} \frac{1}{2} \Gamma(1/2)}$$

$$= \frac{2(\Gamma(1/4))^2}{3\sqrt{\pi}}$$

$$= \frac{8(\Gamma(1/4))^2 \pi^{1/4}}{18 \times 3\sqrt{\pi}}$$

P.T.

$$\beta(m, m) \beta\left(m + \frac{1}{2}, m + \frac{1}{2}\right) = \frac{\pi}{m} 2^{1-4m}$$

$$\text{LHS} = \beta(m, m) \beta\left(m + \frac{1}{2}, m + \frac{1}{2}\right)$$

$$= \frac{\Gamma(m) \Gamma(m)}{\Gamma(2m)} \frac{\Gamma\left(m + \frac{1}{2}\right) \Gamma\left(m + \frac{1}{2}\right)}{\Gamma(2m+1)}$$

$$= \frac{(\Gamma(m) \Gamma\left(m + \frac{1}{2}\right))^2}{\Gamma(2m) 2m \Gamma(2m)}$$

$$= \left[\frac{\Gamma(m) \Gamma\left(m + \frac{1}{2}\right)}{\Gamma(2m)} \right]^2 \frac{1}{2m}$$

$$= \left(\frac{\sqrt{\pi}}{2^{2m-1}} \right)^2 \frac{1}{2m} = \frac{\pi}{2^{4m-2} (2m)}$$

$$= \frac{\pi}{m} 2^{1-4m}$$

$$\Gamma(n) = (n-1) \Gamma(n-1)$$

By Duplication

$$2 \Gamma(m) \Gamma\left(m + \frac{1}{2}\right) = \sqrt{\pi} \Gamma(2m)$$

$$\frac{\Gamma(m) \Gamma\left(m + \frac{1}{2}\right)}{\Gamma(2m)} = \frac{\sqrt{\pi}}{2^{2m-1}}$$